# ECOLOOP: From Waste to Worth

A Student-Led Circular Economy & Digital Platform Initiative



## The 6,000 km Bridge

exchange context

SSDPS

**Swami Sant Dass Public School • Jalandhar, Punjab, India**

Continental urban center tackling high-density packaging and school waste.

Lycée

**Reunion Island High School • French Territory, Indian Ocean**

Island ecosystem collaborating via eTwinning (Belgium, Norway, Czech Republic).

**Core Premise:** Two distant high school communities applying technology and engineering to solve localized waste bottlenecks.

## Presentation Scope

agenda

01 **Physical Raw Material Interception**

Transforming campus packaging into 6 verified functional goods.

02 **The Zero-Cost Digital Marketplace**

React 18 + serverless lead synchronization for $0.00/month.

03 **The Agentic AI Workflow**

Using open-source terminal agents (OpenCode) to ship full-stack code in 48 hours.

# The Physical Foundation: 6 Verified Goods



## Plastics & Packaging

recycled plastics

**01 Eco-Bricks**
Standardized PET bottles packed with clean soft plastics for non-structural garden borders.

**02 MLP Sheets**
Multi-layered plastic wrappers thermally compressed into weather-proof partition boards.

Targets non-recyclable metallized chip/snack packaging.

## Textiles & Cellulose

upcycled materials

**03 Uniform Bags**
Outgrown school uniforms laundered and stitched into heavy-duty carry totes.

**04 Seed Paper**
Post-consumer notebook paper pulped and embedded with native germinable seeds.

Intercepts annual uniform and notebook turnover.

## Biology & Upcycling

organic recovery

**05 Bioenzymes**
Canteen citrus peels fermented over 90 days into non-toxic cleaning solution.

**06 Bottle Planters**
Recovered PET beverage bottles fitted with wicking cords for classroom herbs.

Diverts methane-producing organic waste from dumps.

Strict Criterion: Practical functional utility only. Zero decorative craft items.

# The Challenge: The Science Fair Trap



THE RECURRING FAILURE MODE

**”99% of school environmental projects build great prototypes, win an exhibition certificate, and then rot in a dusty display cabinet forever.”**

Without an active digital inventory, transparent material data, and an inquiry routing pipeline, a physical recycling project cannot sustain itself beyond exhibition day.

# The Platform: Live Web Catalog



## Marketplace Catalog

client layer

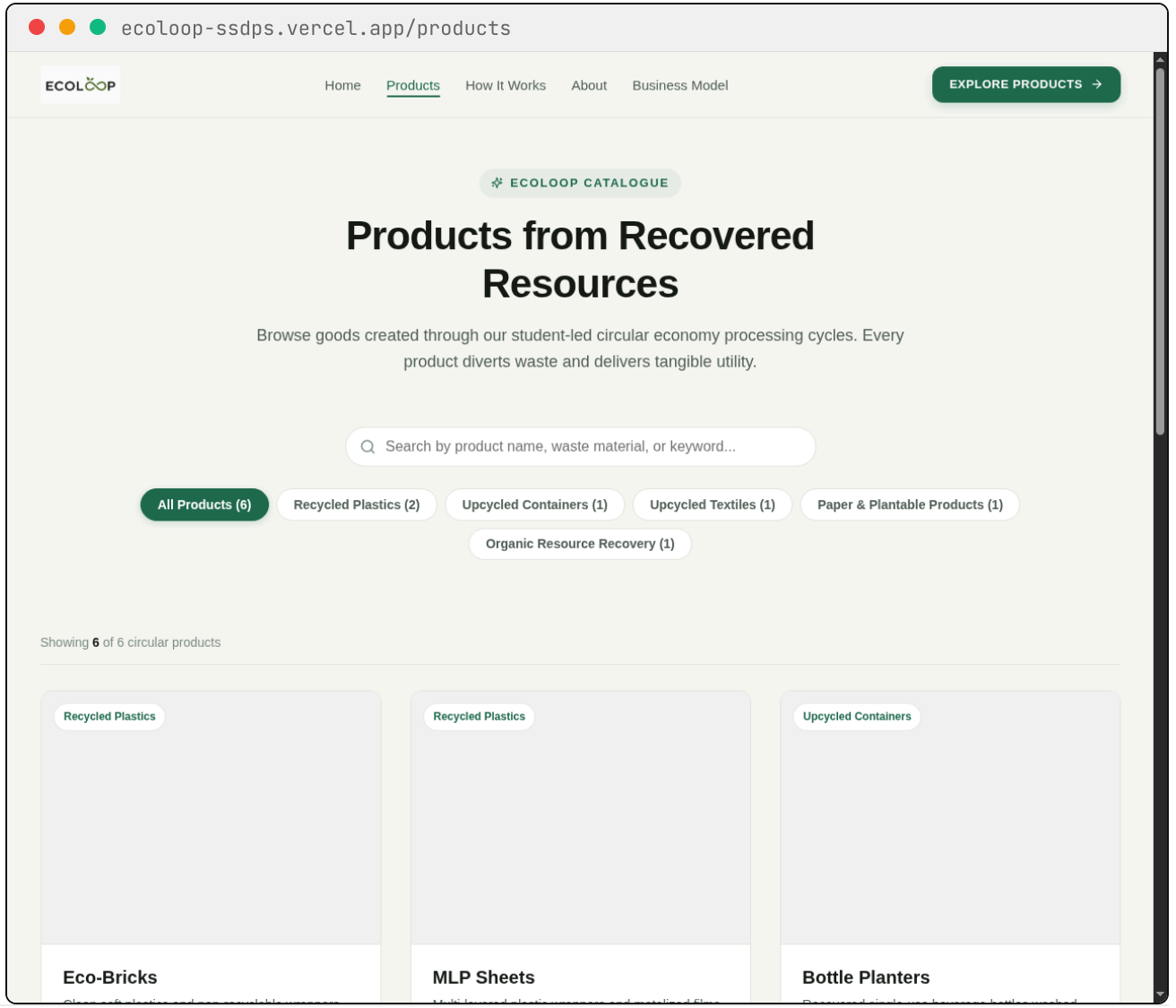
**01 Dynamic Category Filtering**
Instant client-side sorting across Recycled Plastics, Textiles, and Organic Recovery.

**02 Real-Time Query Search**
Searches by input material, product name, or campus application.

**03 Mobile-First Responsive Layout**
Optimized touch interaction for students browsing on smartphones.

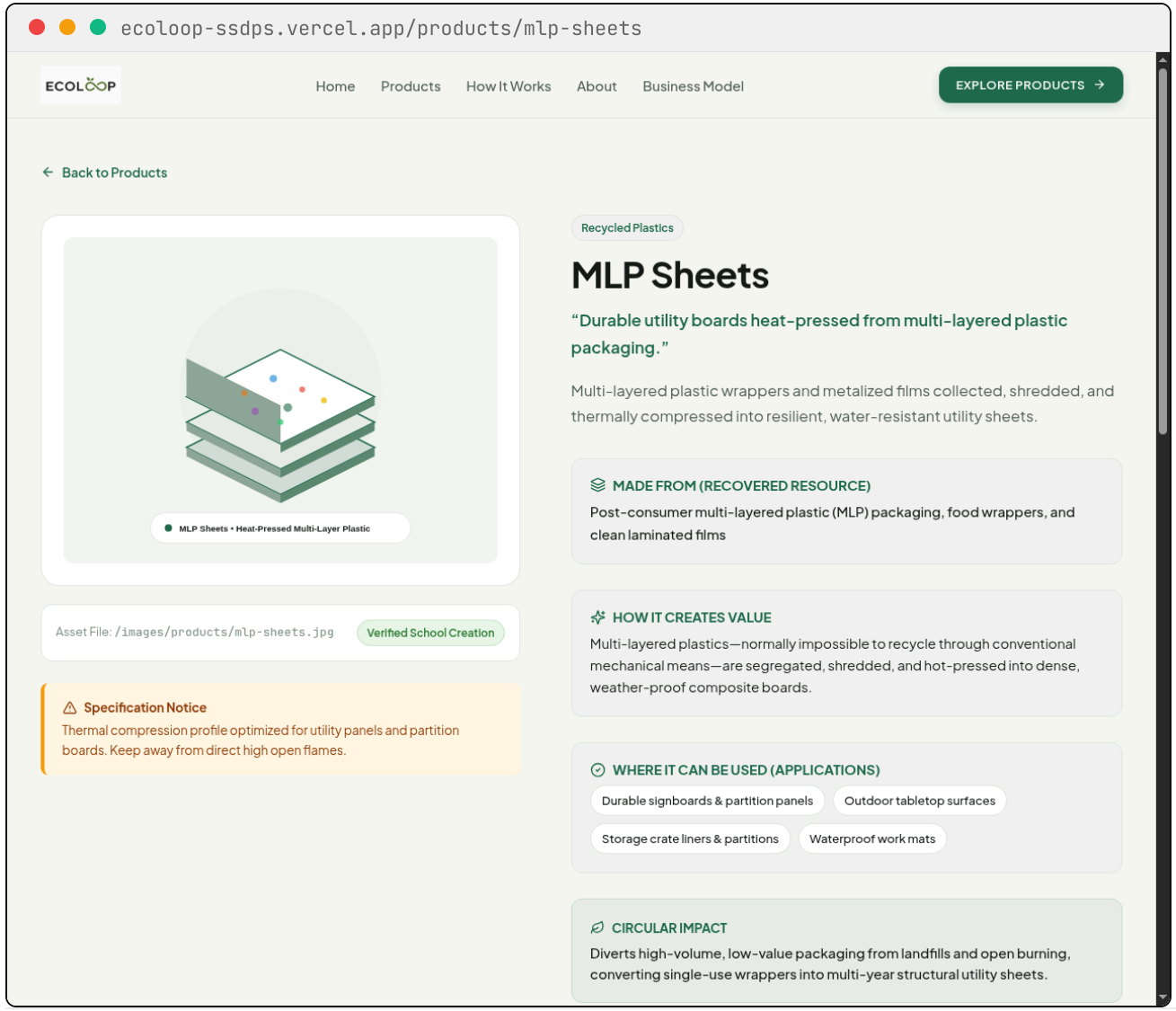
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Built with React 18, TypeScript, and Vite.



# Material Transparency & Honesty






## Rigorous Specifications

verified data

`input` **Recovered Material Sourcing**
Every product explicitly names its input waste stream (e.g. post-consumer MLP packaging).

`caution` **Strict Safety & Caution Notices**
Mandatory disclosures: e.g. Eco-bricks are non-structural; Bioenzymes are non-clinical.

`impact` **Circular Lifecycle Tracking**
Transparent explanation of value creation without exaggerated environmental claims.

Zero greenwashing. Complete engineering honesty.

# The $0.00 Serverless Architecture



## Decoupled Data Pipeline

resilience

**01** **React 18 + TypeScript Frontend**
Single Page Application compiled to static assets and deployed to edge CDN (Vercel).

**02** **Client-Side LocalStorage Buffer**
Caches product inquiries locally so submissions never drop during weak campus Wi-Fi.

**03** **Google Apps Script Serverless Webhook**
HTTP POST endpoint receives payload and authenticates submission with zero server maintenance.

**04** **Google Sheets Live Datastore**
Acts as a real-time collaborative database accessible to faculty and student leads.

## Operating Cost Ledger

budget: 0.00 €

```
// Monthly Infrastructure Invoice
Frontend Edge Hosting: $0.00 (Vercel Tier)
Backend API Computing: $0.00 (Google Apps Script)
Database Storage: $0.00 (Google Sheets)
SSL Certificate & CDN: $0.00 (Included)
-------------------------------------
TOTAL MONTHLY OVERHEAD: $0.00 / month
```

**Takeaway for Students:** You do not need funding or enterprise cloud subscriptions to launch a production-grade digital platform.

# Beyond Chatbots: The Agentic Harness



## 01 / The Consumer Approach

isolated chatbox

Treating AI as an automated essay or text generator:

```
// Typical Student Prompt:
"Write a 400-word paragraph about recycling plastic wrappers
with a strong introduction..."
```

- ✗ **Zero Software Output:** Produces throwaway words, not software.
- ✗ **Zero Engineering Feedback:** No compiler verification, no runtime testing.

## 02 / The Agentic Coding Harness

autonomous loop

Pair-programming with an autonomous software agent:

```
// The Three Pillars of a Harness:
1. Model (The Brain): LLM reasoning engine.
2. Tools (The Hands): Reads files, edits diffs, runs terminal.
3. Environment: Local repo, compiler, linter, and git.
```

- ✓ **Self-Correcting:** Agent reads compiler errors and fixes bugs itself.
- ✓ **Direct Production:** Edits components, compiles bundles, pushes commits.

The Paradigm Shift: From passive prompt-answering to active workspace manipulation.

# The Feedback Loop: How We Shipped Fast



## The 5-Step Autonomous Cycle

execution engine

**01 Specification (AGENTS.md)**
Engineer defines product data, brand constraints, and routing architecture in Markdown.

**02 Autonomous Code Generation**
Agent edits React TypeScript components and updates CSS design tokens.

**03 Local Compiler Execution**
Agent executes `tsc && vite build` directly in the project workspace.

**04 Autonomous Self-Correction**
If TypeScript throws a type mismatch, the agent inspects stderr and repairs the file.

**05 Git Persistence**
Agent stages modified files and commits clean, version-controlled checkpoints.

## Compiler Verification Terminal

production build

```
> ecoloop-marketplace@1.0.0 build
> tsc && vite build

vite v6.4.3 building for production...
✓ 1607 modules transformed.

dist/index.html 1.72 kB | gzip: 0.76 kB
dist/assets/index.css 19.25 kB | gzip: 4.04 kB
dist/assets/index.js 260.57 kB | gzip: 75.94 kB

✓ built in 1.90s with zero errors
```

**Human Role:** Architect, product designer, and quality verifier.
**Agent Role:** Syntax executor, boilerplate writer, and compiler troubleshooter.

# OpenCode: Free & Open Source Agent



## What is OpenCode?

opencode.ai

**01 Open Source Terminal & Desktop Agent**
A fully open-source AI coding harness developed for terminal TUIs and desktop environments.

**02 Zero Financial Requirement**
100% free software. Does not require paid IDE subscriptions or enterprise accounts.

**03 Two Specialized Operating Modes**
`plan` mode explores code safely; `build` mode executes full autonomous edits.

**04 GitHub Actions Integration**
Comment `/opencode` on GitHub PRs and issues to automate cloud coding for free.

## Installation & Setup

1-minute setup

```
# Install OpenCode globally via npm
npm install -g @opencode/cli@latest

# Or install via standalone script
curl -fsSL https://opencode.ai/v2/install | bash

# Launch in any project directory
opencode
```

Runs natively on Linux, macOS, and Windows.

# Running on $0: Free Models & Local LLMs



## Approach 1: Free Cloud Tiers

high speed

**Groq** — **High-Speed Free Inference**

Generous free rate limits running Llama 3 models at 300+ tokens/second.

**Router** — **OpenRouter Free Endpoints**

Access dozens of competitive open-weight coding models with zero credit card entry.

Ideal for laptops with standard specifications and active internet connection.

## Approach 2: Offline Local LLMs (Ollama)

100% offline

**Ollama** — **Local Model Execution**

Run models directly on your hardware without transmitting code over the internet.

**Models** — **Top Free Coding Weights**

`qwen2.5-coder:7b`, `llama3.1:8b`, or `deepseek-coder` run on consumer laptops.

```
ollama run qwen2.5-coder:7b
# Point OpenCode to local Ollama host:
export OPENCODE_PROVIDER=ollama
```

# The Reunion Island Blueprint: 48-Hour Plan



## Step 1: Data Structuring

input

**01 Index Cleanup Volume**

Log kilograms of bottles, wrappers, and beach plastic gathered during cleanups.

**02 Gather Radio Podcast Audio**

Consolidate audio files and student reflection links into a clean Markdown table.

## Step 2: Agentic Scaffolding

build

**03 Initialize OpenCode**

Run `opencode` inside a new Vite web repository.

**04 Autonomous UI Generation**

Prompt agent to scaffold a podcast audio player and live rubbish metric dashboard.

## Step 3: Global Deployment

ship

**05 Deploy to GitHub Pages / Vercel**

Push repo to GitHub; automated actions deploy live site in under 60 seconds.

**06 Public International Hub**

Partner schools in Belgium, Norway, and Czech Republic can stream episodes and view stats.

# Open Discussion & Technical Q&A



## Discussion Prompts

open floor

material

### Plastics & Processing

Questions on multi-layered plastic thermal compression, density, or bioenzyme fermentation.

software

### Digital Architecture

Questions on React 18, Google Apps Script serverless routing, or Google Sheets datastores.

ai agent

### OpenCode & Agentic Workflows

Questions on setting up open-source coding harnesses, local LLMs, or prompt specifications.

## Thank You.

Looking forward to hearing about your local environmental initiatives and software experiments.

Swami Sant Dass Public School • Jalandhar, India